

Adrian Ramlal

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EDUCATION

University of Waterloo

Sep. 2024 – Aug. 2026

Master of Applied Science, Systems Design Engineering

Waterloo, ON

- Thesis: *Physics-Informed Dynamic 3D Reconstruction: From Gaussian Splatting and Rigid-Body Simulation to Inverse Material Estimation*
- Advisor: Prof. John S. Zelek | GPA: 97.75%

University of Toronto

Sep. 2018 – Apr. 2023

Bachelor of Applied Science, Computer Engineering (Honours)

Toronto, ON

- GPA: 3.89/4.00 | Dean's List (2018–2023) | Minors in AI Engineering and Engineering Business

PUBLICATIONS

- [1] **A. Ramlal** and J. S. Zelek, "PersistGS: Differentiable Physics for Object Permanence in 4D Gaussian Splatting," *CVPR Workshop on Generative 3D Reconstruction*, 2026 (Oral). Also presented at the *ICRA Real2Sim/Sim2Real Workshop*, 2026 (non-archival).
- [2] **A. Ramlal**, Y. Chen, and J. S. Zelek, "Latent Space Reinforcement Learning for Inverse Material Estimation in Food Fracture Simulation," *CVPR MetaFood Workshop*, 2026.
- [3] **A. Ramlal** and J. S. Zelek, "Real-Time Physics Simulation with Dynamic Mesh-Gaussian Reconstructions," *Conference on Vision and Intelligent Systems (CVIS)*, 2025 (Oral).
- [4] **A. Ramlal** and J. S. Zelek, "Beyond Static Gaussians: An Empirical Investigation of Architectural Paradigms for Dynamic 3D Scene Reconstruction," *Conference on Vision and Intelligent Systems (CVIS)*, 2025.
- [5] S. Jang, **A. Ramlal**, Y. Chen, and P. W. Fieguth, "Anisotropic Kernels for Neural Implicit Surface Reconstruction," *Conference on Vision and Intelligent Systems (CVIS)*, 2025 (Oral).
- [6] B. Maser, A. Srebrnjak Yang, **A. Ramlal**, and J. S. Zelek, "Optical Flow-Enhanced Thermal UAV Detection Under Camera Ego-Motion for Real-Time Tactical C-UAV," *Conference on Vision and Intelligent Systems (CVIS)*, 2025 (Oral).
- [7] **A. Ramlal**, Y. S. Hu, and J. S. Zelek, "Optimizing 3D Gaussian Splatting via Point Cloud Upsampling," *Conference on Vision and Intelligent Systems (CVIS)*, 2024 (Oral; Best Imaging Paper Award).
- [8] F. J. Pena Cantu, **A. Ramlal**, J. Lei, V. Agrawal, K. Buzko, Y. Xu, A. Wong, J. S. Zelek, Y. Cui, and Y. Chen, "Office Hours: A Multiday Office Cubicle Dataset for Associative Embodied VQA," manuscript, 2026.

RESEARCH EXPERIENCE

Research Associate

Sep. 2026 – Present

Vision and Image Processing Lab, University of Waterloo

Waterloo, ON

- Working with Magna International on knowledge graphs of CAD assemblies: an ontology of system hierarchy, connection types, and functional roles, populated by rule-based geometric pattern matching and LLM-based reasoning.

Graduate Research Assistant

Sep. 2024 – Aug. 2026

Vision and Image Processing Lab, University of Waterloo

Waterloo, ON

- Built PersistGS [1], which keeps fully occluded objects in a 4D Gaussian Splatting reconstruction by estimating friction and initial velocity from the visible frames and simulating the unseen motion with a differentiable rigid-body simulator (NVIDIA Newton/Warp). Modified the simulator's contact kernels so gradients pass through collisions. On three synthetic scenes, PSNR during occlusion is 2.46 dB above constant-velocity extrapolation and within 0.19 dB of using the ground-truth trajectory.
- For inverse material estimation in a non-differentiable MPM fracture simulator (orange peeling) [2], trained a neural surrogate on 2,000 simulations and a goal-conditioned PPO policy in a normalizing-flow latent space. Without per-target retraining, the policy reaches 82% of the recovery of evolutionary search (CMA-ES) with $625 \times$ fewer evaluations.
- Integrated 3D Gaussian Splatting with the Genesis physics simulator [3]. Converting the reconstructions to the fixed-topology meshes that efficient collision detection requires caused 65–80% geometric degradation, which led to the physics-in-the-loop design of PersistGS.
- Proposed a depth-guided point-lifting method and compared it with five upsampling strategies for densifying the sparse COLMAP point clouds that initialize 3D Gaussian Splatting, on 17 scenes [7].

Undergraduate Research (Capstone)

University of Toronto, supervised by Prof. Jon Eyolfson

Sep. 2022 – Apr. 2023

Toronto, ON

- Designed an LSTM for controllable music generation, conditioned on harmonic and rhythmic parameters, with custom melodic-quality and timbral-similarity metrics in the loss (Certificate of Distinction).

AWARDS AND FELLOWSHIPS

Engineering Graduate Merit Award, University of Waterloo	2026
Ontario Graduate Fellowship (Engineering), University of Waterloo	2025
University of Waterloo Graduate Scholarship	2025
International Master’s Award of Excellence, University of Waterloo	2024–2026
Best Imaging Paper Award, Conference on Vision and Intelligent Systems (CVIS)	2024
David Johnston International Student Entrance Scholarship, University of Waterloo	2024
Engineering International Scholar Award, University of Toronto	2018–2023
Trinidad and Tobago National Open Scholarship (top 1% nationally)	2018

INDUSTRY EXPERIENCE

Full-Stack Software Developer

Ministry of Education

Sep. 2023 – Aug. 2024

Port of Spain, Trinidad and Tobago

- Built role-based access control with per-page permissions and audit logging for a national education platform.
- Integrated the arXiv API to automate article ingestion and metadata extraction.

Digital Logic Design Intern

Intel Corporation

May 2021 – Aug. 2022

North York, ON

- Developed SystemVerilog behavioural models and testbenches for high-speed multi-protocol serial deserializers in Intel’s chip-to-chip interconnect products.
- Automated the translation of analog schematics into behavioural models.
- Validated the mixed-signal models against silicon measurements, working with the RTL, analog, and verification teams.

TEACHING EXPERIENCE

Teaching Assistant

University of Waterloo

Sep. 2025 – Apr. 2026

Waterloo, ON

- Pattern Recognition (SYDE 675), Winter 2026
- Deep Learning (SYDE 577), Fall 2025

TECHNICAL SKILLS

Languages: Python, C/C++, CUDA, Rust, MATLAB, SystemVerilog

Software: PyTorch, JAX, NVIDIA Newton/Warp, Genesis, Isaac Lab, nvdiffrast, COLMAP, Open3D, OpenCV, OpenGL, Blender